

$$\begin{aligned}\int_E f(x)g(x)dx &= \int_E f^l(x)f^{1-l}(x)g(x)dx \\ &\leq \left(\int_E f^{ls}(x)dx\right)^{1/s} \left(f^{(1-l)s'}(x)g^{s'}(x)\right)^{1/s'}.\end{aligned}$$

令 $ls = p$, $\frac{1}{s} = \frac{1}{p} - \frac{1}{r} = 1 - \frac{1}{q}$, 因而有 $s' = q$, 于是

$$\begin{aligned}\int_E f(x)g(x)dx &\leq \|f\|_p^{1-p/r} \left(\int_E f^{pq/r}(x)g^q(x)dx\right)^{1/q}, \\ \left(\int_E f^{pq/r}(x)g^q(x)dx\right)^{1/q} &= \left(\int_E f^{pq/r}(x)g^{q-l}(x)g^l(x)dx\right)^{1/q} \\ &\leq \left(\int_E g^{ls}(x)dx\right)^{1/sq} \left(\int_E f^{pq s'/r}(x)g^{(q-l)s'}(x)dx\right)^{1/s'q}.\end{aligned}$$

令 $ls = q$, $\frac{1}{sq} = \frac{1}{q} - \frac{1}{r}$, 因而有 $\frac{1}{s} = 1 - \frac{q}{r}$, $s' = r/q$, $l = q/s = q - q^2/r$. 于是

$$\text{有 } \int_E f(x)g(x)dx \leq \|f\|_p^{1-p/r} \|g\|^{1-q/r} \left(\int_E f^p(x)g^q(x)dx\right)^{1/r}.$$

10. 由叶戈罗夫定理, $\forall \delta > 0, \exists E_\delta \subset E = [a, b]$, 使得 $m(E_\delta) < \delta$, 在 $E \setminus E_\delta$ 上 $f_k \Rightarrow f$. 因为

$$f_k \in L^r(E) \subset L^p(E), \quad \int_E |f|^r dx \leq \lim_{k \rightarrow \infty} \int_E |f_k|^r dx \leq M,$$

所以 $f \in L^r(E) \subset L^p(E)$. 又

$$\begin{aligned}\int_E |f_k - f|^p dx &= \int_{E_\delta} |f_k - f|^p dx + \int_{E \setminus E_\delta} |f_k - f|^p dx \\ &\leq 2^p \int_{E_\delta} (|f_k|^p + |f|^p) dx + \int_{E \setminus E_\delta} |f_k - f|^p dx \\ &\leq 2^p \left[\left(\int_{E_\delta} |f_k|^r dx \right)^{p/r} + \left(\int_{E_\delta} |f|^r dx \right)^{p/r} \right] m(E_\delta)^{1-p/r} \\ &\quad + \int_{E \setminus E_\delta} |f_k - f|^p dx, \\ \lim_{k \rightarrow \infty} \int_E |f_k - f|^p dx &\leq 2^p M m(E_\delta)^{1-p/r} < 2^p M \delta^{1-p/r}.\end{aligned}$$

令 $\delta \rightarrow 0$ 即得.